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RESEARCH

Simplex Projection Predictions of the Remainder of Solar Cycle 25 and the Next Solar Cycle 26 Based on the Monthly Mean Sunspot Numbers

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Abstract
Solar activity and related space weather have a significant impact on the Solar System and Earth. Therefore, reliable prediction of solar activity is becoming increasingly important. To predict the remainder of Solar Cycle 25 and the amplitude and timing of the maximum of Solar Cycle 26, we applied the Simplex Projection method to the monthly mean sunspot number (SSN). While most prediction studies rely on 13-month smoothed SSN, we deliberately used unsmoothed monthly mean SSN for long-term solar cycle postcasts and obtained quite successful single and double cycle predictions for Solar Cycles 20 – 24. We defined the "split point" between the library and prediction sets as an important setting parameter; when this parameter was optimized, prediction ability improved significantly. We predicted that Solar Cycle 26 will be slightly weaker than Solar Cycle 25 and stronger than Solar Cycle 24. Its cycle profile is expected to show either a well-defined double peak or a slightly fluctuating flat peak, both resembling Solar Cycle 20. We predicted that the Solar Cycle 25 minimum will occur in the mid-2030. For Solar Cycle 26, the maximum is predicted for June 2035 with 150.6 – 181.5 monthly mean SSN (13-month smoothed 137.4 – 146.2), and the minimum for late 2040. The similarity between Solar Cycles 20 and 26 may reflect Gleissberg Cycle modulation.

Keywords Solar cycle, models · Sunspots, statistics

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1. Introduction

The Sun exhibits a quasi-periodic behavior and can be considered as an active star. It is known that there are many active stars, but one thing that sets the Sun apart is its proximity to the Earth. Given this proximity, small variations in solar activity become significant, making it the most important and influential star for the Earth and life on it. The reason is that solar activity variations affect not only the space weather and the interplanetary environment but also directly and indirectly affect the Earth and life on it. Therefore, it is valuable to be able to make long-term and short-term predictions of solar activity to improve our understanding of solar activity and its effects.

The progress and development of our civilization over the last century has been driven by the use of technological and electronic systems. A sudden and small change in solar activity may cause a significant impact on the performance of these systems, therefore, understanding and prediction of solar activity is of high priority. An important factor in investigating and understanding of the Sun–Earth relationship is the analysis and prediction of space weather resulting from various solar activity events. Space weather effects arise from dynamic and highly variable conditions in the geospace environment, starting from explosive events observed on the Sun, such as solar flares and coronal mass ejections, and extending to various energetic effects in the magnetosphere–ionosphere–atmosphere system (Singh, Siingh, and Singh 2010). Studying the nonlinear and chaotic nature of these effects and their relationships, along with the relevant parameters, is crucial to improve future predictions. Therefore, accurate prediction of solar activity indices within the studied dynamical system provides a better understanding of the impact of space weather on Earth and life.

Solar activity stems from the Sun's magnetic field (e.g. Svalgaard 2013). Monitoring and analysis of changes in solar activity is conducted by examining solar events and the parameters that characterize them. Various solar activity indicators, such as sunspot numbers (SSN), sunspot areas (SA), solar flares (SF), the 10.7 cm solar radio flux (F10.7), and coronal mass ejections (CMEs) are used to monitor and analyze the temporal behavior of solar activity, as well as to predict future solar activity (Hathaway, Wilson, and Reichmann 1994; Sarp et al. 2018; Pesnell and Schatten 2018; Wang et al. 2020; Kilcik et al. 2020; Chowdhury et al. 2022; Zhu, Zhu, and He 2022; Gopalswamy et al. 2023). Among these, one of the best indicators of solar variability is the number of sunspots visible on the solar disk (Echer et al. 2004). Records of observed sunspot numbers reveal a quasi-periodic cycle of approximately 11 years (Eddy 1976).

Studies have been conducted to understand the structure and features of solar cycles and to unravel their chaotic dynamics. Earlier studies, such as Jevtić, Schweitzer, and Cellucci (2001), used phase-space reconstruction techniques to identify a correlation dimension indicative of chaotic dynamics. Letellier et al. (2006) supported this finding with nonlinear time series analysis, suggesting the presence of low-dimensional chaos in solar activity. Spiegel (2009) further confirmed the presence of deterministic chaos through methods like Lyapunov exponents and fractal dimensions. Kilcik et al. (2009) and Crosson and Binder (2009) expanded this understanding by investigating nonlinear prediction methods and applying chaos theory to solar cycle predictions. Recent advancements in chaos theory and machine learning (ML) have introduced new perspectives on solar cycle prediction. McIntosh and Leamon (2017), McIntosh et al. (2020) proposed a new approach using empirical relationships between solar magnetic proxies (bright points (BPs), g-node density and solar magnetic field) to predict Solar Cycle 25, highlighting the importance of chaotic interactions among magnetic fields. Petrovay (2020) reviewed various nonlinear dynamo models and their capability to detect solar cycle variations, offering deeper insights into the mechanisms driving solar activity. These studies increasingly incorporate nonlinear dynamics to better simulate and predict the chaotic fluctuations observed in sunspot cycles. The application of chaos theory and nonlinear dynamics has thus gained significant importance, providing valuable information for understanding the complexity of solar cycles and enhancing long-term prediction models. Ongoing research continues to explore new methodologies for modeling the interplay between order and chaos in solar data, which will likely lead to more accurate predictions of future solar activity.

While solar activity may appear to exhibit a degree of order or linearity, there exists a hidden chaos within this apparent order, as well as an underlying order within the chaos. This phenomenon is known as "deterministic chaos". Consequently, it is reasonable to attempt predictions of the dynamic and chaotic parameters of solar activity based on their linear characteristics. Nonlinear methods have demonstrated their effectiveness in modeling chaotic systems (Sugihara 1994; Maye et al. 2007; Tsonis et al. 2015). In particular, the Empirical Dynamic Modeling (EDM) approach and the Simplex Projection method derived from EDM have achieved notable success in such studies (Sugihara and May 1990; Sugihara et al. 2012; Deyle et al. 2013; Ye et al. 2015). EDM and Simplex Projection methods offer a model-free approach for detecting the nonlinear interactions among variables within complex systems, making it a suitable choice for predicting solar activity.

Growing evidence indicates that solar cyclic variability is substantially influenced (if not dominated) by stochastic perturbations (Bhowmik et al. 2023; Wang, Jiang, and Wang 2025). It is widely characterized as a stochastically forced deterministic system, such that both nonlinear dynamics and stochastic perturbations jointly shape the amplitude and phase of solar activity (Charbonneau and Dikpati 2000; Cameron et al. 2014; Lemerle, Charbonneau, and Carignan-Dugas 2015; Charbonneau 2020; Jiang 2023). Here, a completely deterministic mechanism is not assumed; instead, the EDM and Simplex Projection methods are used as model-free, data-driven prediction tools. The results are interpreted as statistical predictions and it is expected that the prediction skill will decrease, especially with increasing stochasticity over longer time periods (Sugihara and May 1990; Sugihara 1994).

In our study, the Simplex Projection method is applied to monthly mean sunspot number data to provide predictions of Solar Cycle 25 and the next Solar Cycle 26. Single and double postcasts for past cycles (Solar Cycles 20 to 24) are also included to test the performance of the prediction method. Section 2 describes the data used in the study and discusses the methods and analysis. Section 3 presents the results of the study, the postcasts and the predictions. Section 4 demonstrates the sensitivity of prediction performance to the split point between the library and the prediction sets, and finally, Section 5 presents the discussion and conclusions.

2. Data and Methods

The data used in this study are the monthly mean total sunspot number and the 13-month smoothed monthly total sunspot number provided by the World Data Center SILSO, Royal Observatory of Belgium, Brussels (SILSO, World Data Center - Sunspot Number and Long-term Solar Observations, Royal Observatory of Belgium, on-line sunspot number catalog: http://www.sidc.be/SILSO/home, 1749 – 2024). For brevity's sake we will refer to them as monthly mean and monthly smoothed data sets. The monthly mean data cover the period from January 1749 to December 2024, while the monthly smoothed data cover the period from July 1749 to June 2024.

Here, we used the Simplex Projection method which is part of Sugihara's Empirical Dynamic Modeling (EDM) for the analysis of nonlinear and chaotic systems. The method is specifically designed to analyze time series data by using mathematical and geometric techniques to predict the future state from the past state of the data. The aim of EDM, and by extension Simplex Projection, is to understand, model and predict the behavior of systems by reconstructing system dynamics from time series data. Time series are sequential observations of system behavior, and therefore information about the rules governing system behavior (i.e., system dynamics) is hidden/embedded in the data. EDM/Simplex is equation-free (model-free): it reconstructs system dynamics from time-delay embeddings of observed time series without specifying the governing equations. The prediction skill is generally highest when the series exhibits a state-dependent (nonlinear) deterministic structure, and decreases as stochastic components become dominant; thus, the method provides statistical rather than mechanical predictions (Sugihara and May 1990; Sugihara 1994; Ye et al. 2015). For the solar cycle, increasing stochasticity can further limit predictability (Bhowmik et al. 2023; Wang, Jiang, and Wang 2025). Takens' Theorem expresses a way to recover this information using only a single time series (Takens 1981):

$$X(t) = [x(t), x(t-\tau), x(t-2\tau), \ldots, x(t-(E-1)\tau)], \tag{1}$$

where X is a vector of the original state-space, x is a sample in the time series t, τ (tau) is a time delay, and E is the embedding dimension used to reconstruct the phase-space. The approximation constructs a vector X(t) that includes the current observation x(t) and its previous values at intervals defined by the time delay τ. By utilizing multiple lagged observations, this method enables the reconstruction of the underlying dynamics of the system from a single time series. The theorem suggests that, under certain conditions, the dynamics of a system can be accurately represented in a higher-dimensional space, allowing for the analysis and prediction of future states based on past data. Therefore, the choice of E and τ is critical to effectively determine the past behavior of the system and ensure that the reconstructed attractor accurately reflects the true dynamics of the underlying process.

The Simplex Projection is a geometric approach used to predict the future state of a system based on its past behavior. Suppose we have a current state vector, X(t), and we want to predict the next state, X(t + 1). First, we first find E + 1 nearest neighbors of X(t) in the embedding space. These neighbors are denoted as $X_1, X_2, \ldots, X_{E+1}$. The predicted values $\mathrm{Pred}_{[1,2,\ldots,T_a]}$ are then calculated as a weighted average of the future values of these neighbors:

$$Pred_{[1,2,\ldots,T_a]} = \sum_{j=1}^{E+1} \omega_i v^1_{(i+[1,2,\ldots,T_a])}, \tag{2}$$

where ω_i are the weights assigned to each neighbor based on their Euclidean distance from X(t). The weights are calculated as:

$$\omega_i = \frac{1}{\sum_{j=1}^{E+1} \exp\left(\frac{-\|v_n - v_j\|}{\omega_{avg}}\right)} \exp\left(\frac{-\|v_n - v_i\|}{\omega_{avg}}\right), \tag{3}$$

where $\|v_n - v_i\|$ is the distance between v_n and v_i, and ω_{avg} is the average scaling factor. The method allows us to predict future states of the system based on its past behavior, as detected in time series data. By iterating this process, we can produce predictions for multiple steps into the future.

The process of embedding time series data into a phase-space is accomplished using Takens' Embedding Theorem, which allows the phase-space of a dynamic system to be reconstructed and represented based on time series data. The theorem provides a framework for representing the dynamics of the system in a higher-dimensional space and detecting key features of the underlying processes. Once the phase-space is created, vectors representing the state of the system at different times are generated. These vectors are constructed using a specified embedding dimension, E, and time delay, τ, creating what is known as the embedding space. Each vector includes past observations up to the embedding dimension, enabling a comprehensive representation of the past behavior of the system.

To find the nearest neighbors and detect similarities, the phase-space data is split into two sets: the library set and the prediction set. The library set consists of past/historical data used to construct the embedding space and to identify nearest neighbors, providing the basis for understanding the system's behavior and its past states. The prediction set, on the other hand, includes the current state points/vectors, which are the data points from which future states are to be predicted. Simplex Projection uses the library set to determine the most similar past states to the points in the prediction set, allowing for accurate predictions based on the identified nearest neighbors. The point/state where the library set ends and the prediction set starts, i.e., where the sets are exactly split (referred to as the split point), is crucial for prediction performance and should be chosen precisely.

During the next step the nearest neighbors of the current state vector are identified within the embedding space. These neighbors are past states that are similar to the current state, and similarity is determined by calculating the distances between the current state vector and all other vectors in the embedding space. The vectors closest to the current state represent the most similar past states, which are critical for accurate predictions. A projection (simplex) is then created using the identified nearest neighbors. This involves calculating a weighted average of the future values associated with these similar neighbors, with the weights being based on the distances between the current state and the neighbors, closer neighbors receive higher weights. This weighted average is used to generate predictions for future states of the system, effectively leveraging the information from the most relevant past data.

The Simplex Projection (SP) method has several advantages compared to precursor method, Autoregressive models (AR), Artificial Neural Networks (ANN), parametric approaches (e.g. ARIMA/SARIMA), Support Vector Regression (SVR) and machine learning (ML) methods (e.g. Random Forests (RF) and Long Short-Term Memory Networks (LSTM)). Unlike AR/ARIMA/SARIMA, it does not assume linearity or stationarity, requires no differencing or order selection, and naturally captures state-dependent dynamics. Compared with ANN/LSTM and SVR, SP achieves nonlinear prediction with minimal hyperparameter tuning (no kernel/architecture choice) and performs reliably on limited, noisy data (i.e. solar cycle data). Unlike RF, which does not preserve temporal order and often requires feature engineering, SP takes into account the phase space geometry by using the nearest neighbors in the reconstructed state space. It is also computationally light: local linear fits avoid global optimization, so training is much easier than deep learning or SVR. SP searches past states in the data sets for close analogs and uses them for prediction. By contrast, polar-field precursors infer the next-cycle amplitude from the Sun's axial-dipole field measured near the preceding minimum; they therefore become informative only late in the cycle, and are not designed to provide within-cycle timing (maxima/minima), double-peak structure, or rolling updates far from minimum. SP can be run at any cycle phase to provide rolling predictions of amplitude, timing (maxima/minima), and the overall cycle profile, with transparent, inspectable analogs.

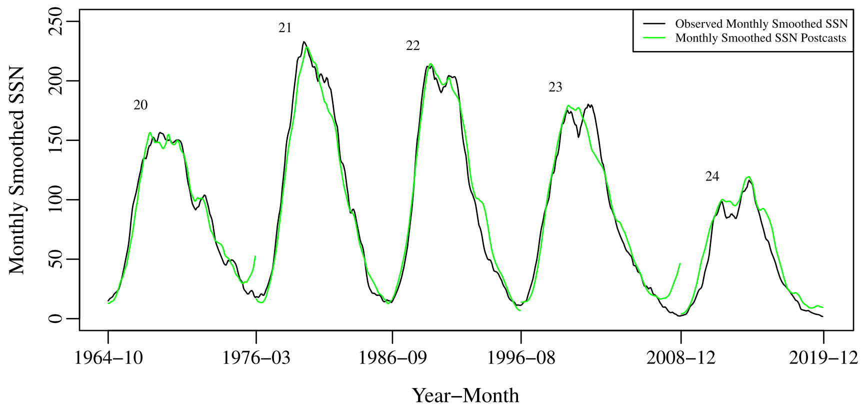

Figure 1 Single postcasts of past solar cycles generated using monthly smoothed sunspot numbers.

3. Analysis and Results

3.1. Single and Double Postcasts of Past Solar Cycles

In this study, the last five solar cycles, Solar Cycles 20 to 24, were postcasted to single cycles (one cycle period) and double cycles (two cycle period) to demonstrate the usability and performance of the prediction method before starting predictions of the remainder of Solar Cycle 25 and the next Solar Cycle 26. For each relevant cycle to be postcast, the observed data set is taken until the end of the previous cycle and the subsequent cycles are postcast. For example, to postcast Solar Cycle 24, the data set was taken until the end of Solar Cycle 23. To determine successful postcasts, mean absolute error (MAE) and correlation coefficient ρ (rho) were used. The calculations for MAE and ρ were performed by comparing all observed values of the relevant single or double cycles with the predictions generated for the postcast. Postcasts were considered successful if ρ values were higher than 0.95 for both data sets. The MAE value was considered successful if it was below 5% for the monthly smoothed data and below 20% for the monthly mean data. Multiple successful postcasts were produced for varying embedding dimension (E) and time delay (τ) (E $=$ 2 to 10 and $\tau = 1$ to 70). As a result, for each embedding dimension, all successful postcasts were analyzed and the highest performance was selected (Figure 1 and Figure 2 for single postcasts of past cycles). In addition to the single cycle postcast, we also postcasted double cycles by using only monthly mean data (Figure 3 for double postcasts of past cycles and averaged versions in Figure 4). As in the single cycle case, the data sets were taken until just before the beginning of the relevant cycle for which the predictions were to be initiated and the postcasts were produced. Successful predictions were selected in a similar way to the single cycle approach. The only difference is that MAE and ρ are calculated over two cycles, i.e., over a longer prediction period. To show the smoothed version of the postcasts/predictions the results were then averaged using a 13-month running average, and we will refer to the postcasted/predicted data set as monthly averaged data. Thus, the long-term prediction performance of the method was demonstrated.

Figure 1 plots the postcasts based on the monthly smoothed data set. Notable correspondence is observed between the monthly smoothed observed (black) and predicted (green) data for cycle maxima, except for the second peak of Solar Cycle 23, which was not properly reconstructed. This can be explained by the fact that Solar Cycle 23 was relatively

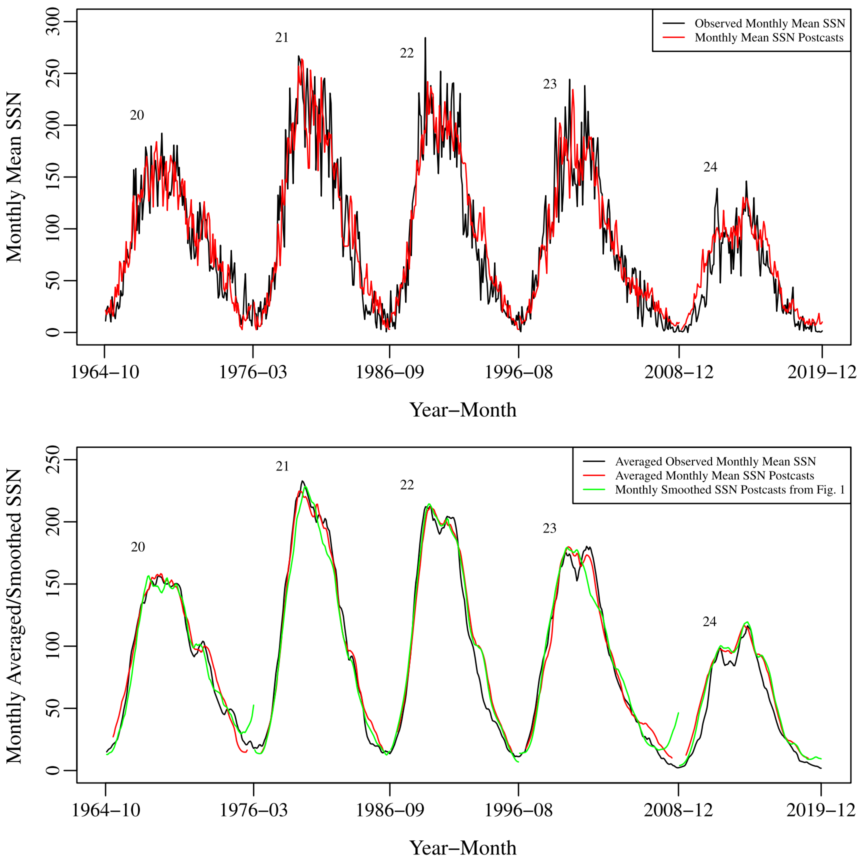

Figure 2 Original (top) and monthly averaged (bottom) version of single postcasts of past solar cycles generated using monthly mean sunspot numbers. The green lines (bottom) show the monthly smoothed single postcasts from Figure 1.

unusual. In particular, it has been referred to as an anomalous cycle characterized by lower sunspot activity compared with Solar Cycles 21 – 22, changes in sunspot number, reduced TSI variations, and unique deviations from previous cycles (de Toma et al. 2004). Additionally, there are significant deviations in the minima for Solar Cycle 23, as well as for Solar Cycle 20. Therefore, the performance of the postcasts generated from the monthly smoothed data is lower in the minimum periods compared to the maximum periods.

Figure 2 top panel shows postcasts based on the monthly mean data set (red), which are also quite consistent with the observed values (black). Monthly mean data appears more chaotic as it directly reflects the short-term variations such as sudden increases and decreases in solar activity. In such data, disturbances and noise components are more pronounced. To reduce small fluctuations and reveal the overall trend in the predicted data we applied a 13-month running average filtering and we will refer to this data set as monthly averaged postcasts/predictions data. As seen in the bottom panel of Figure 2, using monthly mean data for solar cycle predictions produces more accurate results, especially for the cycle maxima, compared to postcasts based on monthly smoothed data (green). The difference is even more prominent for cycle minima. However, it should be noted that postcasts pro-

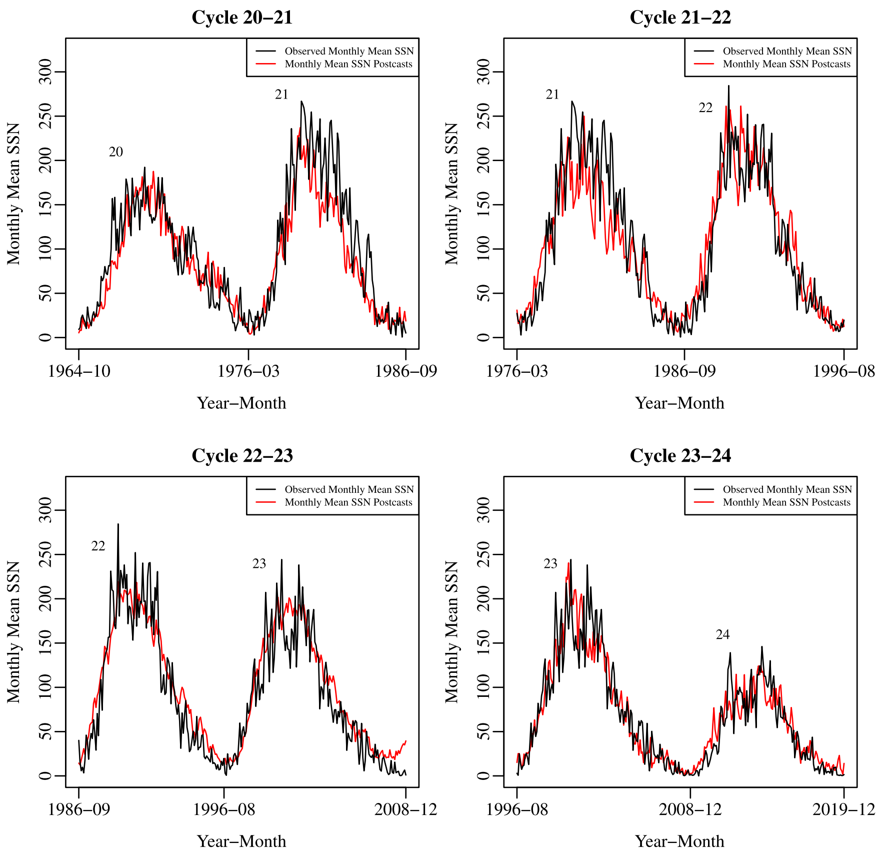

Figure 3 Double postcasts of past solar cycles generated using monthly mean sunspot numbers.

duced from monthly smoothed data do not involve data loss. Monthly averaged postcast data, on the other hand, loses the first and the last six data points due to the edge effect during the filtering. Therefore, for monthly averaged postcasts data (bottom red), there are small gaps at the start and end minima of the cycles. Nevertheless, monthly mean postcast data and average of their results performed much better than the monthly smoothed postcast data (green) for cycle minima. Furthermore, based on the mean correlation coefficient, ρ, we concluded that the monthly averaged postcasts ($\rho = 0.9959$) outperformed monthly smoothed postcasts ($\rho = 0.9952$). Henceforth, to generate postcasts and predictions we will only use the observed monthly mean data that will generate monthly mean predicted and monthly averaged predicted data.

To evaluate outcome and prediction performance at long-term time steps (two cycles) we applied the same method to past five cycles. For these postcasts, we used only the monthly mean SSN, considering the above inference, and generated postcasts for four pairs of solar cycles: 20 – 21, 21 – 22, 22 – 23, and 23 – 24. We present the results in Figure 3 which shows that the amplitude and the overall structure of the solar cycles are well captured. However, some small differences can be observed, especially during the maxima of certain cycles (Solar Cycles 21 and 23), where the postcast values deviate slightly from the observed data. As these deviations are not seen in the single cycle postcasts, we

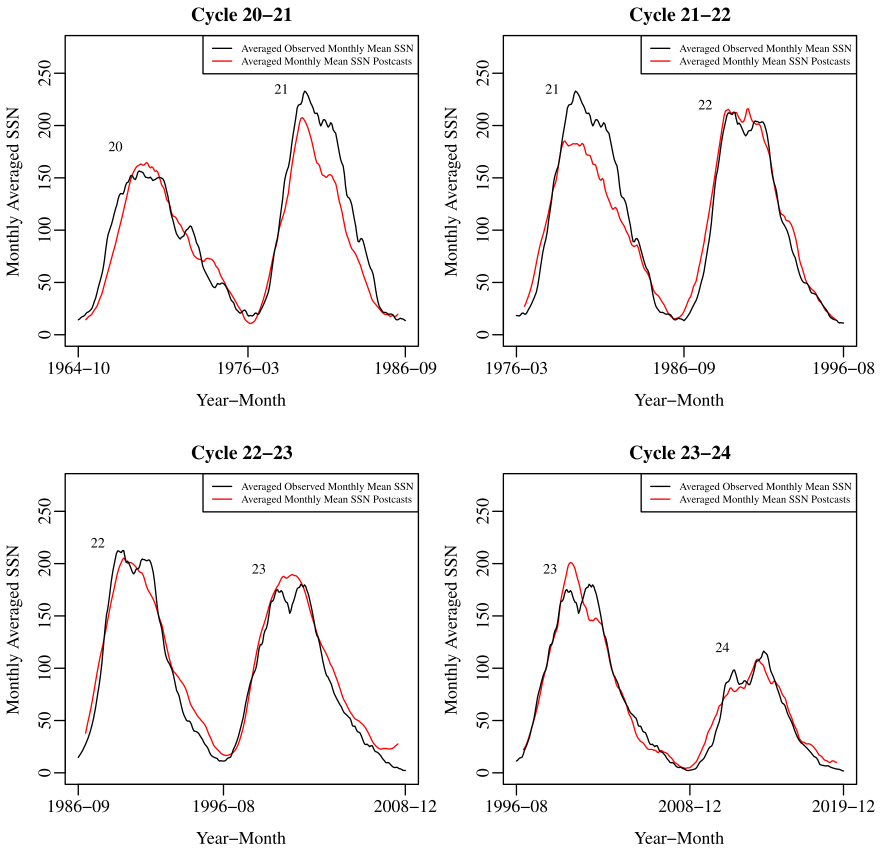

Figure 4 Monthly averaged version of the double postcasts generated using monthly mean sunspot numbers from Figure 3.

attribute them to a consequence of generating long-term predictions, which further suggests that generating long-term predictions for cycles with strong (such as Solar Cycle 21) or well-defined double peaks (such as Solar Cycle 23) might be less accurate. Nevertheless, these deviations in maxima are not excessive and the minima are successfully matched. Thus, the fact that the postcasts produced at long-term time steps using monthly mean data are quite successful, further confirmed successful performance of the prediction method.

Next, a 13-month average was applied to the data presented in Figure 3 and the results are shown in Figure 4. The averaged results shows more clearly the success of double cycle postcasts and the variations compared to Figure 3. Only for the Solar Cycle 21 – 22 pair, the deviation in Solar Cycle 21 is somewhat high, while the overall structure and Solar Cycle 22 are well captured. For Solar Cycles 20 – 21, the declining phase of Solar Cycle 21 is well predicted with a small-scale deviation, which is likely due to the fact that this cycle is stronger than the other cycles analyzed. When considering the Solar Cycle 23 – 24 pair, we attribute the deviation at the peak of Solar Cycle 23 to the fact that it has well-defined double peaks as previously mentioned. In the postcasts for the other double cycles, the amplitudes and structures match quite well and the method proves its success in double cycle prediction.

It can be seen that for all figures and results, the single postcast/prediction performance is better than the double cycle performance, which is an expected result: prediction success for chaotic and nonlinear systems decreases as the prediction time increases (Kantz and Schreiber 2003; Bradley and Kantz 2015; Lorenz 2017). However, the performance did not decrease by a double factor, even though there was double the time step difference between the two approaches. The method also performed well on double cycle postcasts, accurately capturing and predicting almost all double cycles and their amplitudes. Based on these results, we provide a basis for the combined prediction of the remainder of Solar Cycle 25 and the next Solar Cycle 26 and emphasize the feasibility of double cycles prediction.

3.2. Predictions of the Remainder of Solar Cycle 25 and the Next Solar Cycle 26

In order to obtain the most successful predictions of Solar Cycles 25 and 26, we tried all different combinations of split points, E, and τ parameters, which is an important approach to find successful predictions. Therefore, here we focus on the definition of the embedding dimension, E, and the time delay, τ. The embedding dimension determines the number of variables required to reconstruct the system's state-space and is essential for capturing the underlying dynamics of the solar cycle. A higher embedding dimension typically provides more detailed reconstructions of the system's behavior, while lower dimensions might oversimplify it. The time delay represents the temporal spacing between consecutive points used in the state-space reconstruction and helps to capture the autocorrelations in the data. By adjusting these two parameters and split points, we aimed not only to produce the best prediction, but also to test the stability and accuracy of the predictions. Different values of these parameters can lead to slight shifts in the prediction curves, particularly in terms of the time and amplitude of the solar cycle. For instance, in some cases, higher E values may capture more complex dynamics, resulting in slightly earlier or later time of the maxima, while varying τ may affect the smoothness of the transitions during the ascending or descending phases of a solar cycle. The differences observed in the predictions in this study are mainly attributable to the different combinations of split points, E, and τ values used. This approach allows us to show the parameter sensitivity of the predictions, providing a more comprehensive view of the potential shape of Solar Cycles 25 and 26.

For the combined prediction of the remainder of Solar Cycle 25 and the next Solar Cycle 26, we used the monthly mean sunspot number. The data set was taken until the end of Solar Cycle 24 (December 2019) and predictions were performed for two cycles (about 300 months). Predictions were produced for each scenario for varying values of E (from 2 to 10) and τ (from 1 to 70) and varying split points (from 1000 to 2700). To try various combinations of these three parameters, we chose the following approach. Starting with the 1000th split point, the value of the embedded dimension was fixed at $E = 2$ and predictions were produced for each τ. The same calculations were repeated for the 1001st split point and so on until the 2700th split point. After that embedded dimension value was fixed at $E = 3$. We repeated the calculations until $E = 10$. MAE and ρ criteria were again applied to select the most successful predictions. For the predictions these two criteria were calculated based on all observed data points (60 months) of Solar Cycle 25 up to December 2024. Then, we chose the three most successful predictions.

Figure 5 top panel shows the predictions of Solar Cycles 25 and 26 produced using the monthly mean sunspot number, while the bottom panel shows 13-month averaged versions of the same predicted data. The original predictions in the top panel show the three most successful predictions (shown in different colors) that we produced using different parameters for the combined prediction of Solar Cycles 25 and 26. The prediction results and

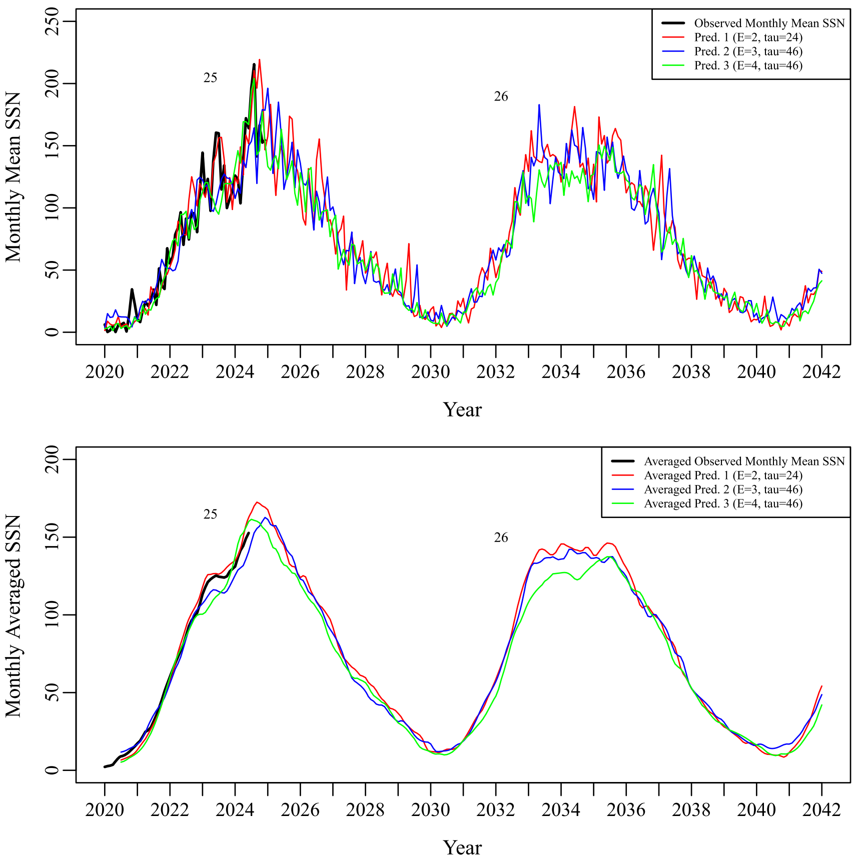

Figure 5 Original (top) and monthly averaged (bottom) version of predictions of Solar Cycles 25 and 26 generated using monthly mean sunspot numbers.

performance during the first half of Solar Cycle 25, the beginning of the predictions, are very important, because all the currently available observed data (black lines) cover this part. The match between the predictions and the observed test points (black lines) allows the most successful predictions to be selected. Hence, the combination of parameters that contain the most probable unknown/future values and potential shapes for the cycles is determined. The bottom panel shows the cycle shapes obtained by applying a 13-month average to the original predictions. Here, the matching of the predictions with the observed part and the potential shapes of the predicted cycles become more apparent. The predictions are quite parallel to each other, with only the Pred. 3 (green) is slightly lower in overall amplitude, especially for Solar Cycle 26.

Solar Cycle 25 has already reached its maximum that has a double peak structure. Our predictions show that Solar Cycle 26 will be weaker than Solar Cycle 25 but stronger than Solar Cycle 24. Overall structure of Solar Cycle 26 could be a well-defined double peak, but more accurate predictions can be made at the end of Solar Cycle 25, when more data points are available. Moreover, the predicted Solar Cycle 26 appears to be similar to Solar Cycle 20 when compared to past cycles. This period of 6 – 7 cycles could be another evidence of the long-term Gleissberg Cycle (Gleissberg 1939), which has a period approximately in

Table 1 Prediction results of the minimum of Solar Cycle 25 and monthly averaged version of the prediction results.

Predictions	Mean Absolute Error (MAE)	Corr. Coeff. (ρ)	Amp. of Min. (SSN)	Time of Min. (yyyy-mm)	Cycle Length (years)
Predictions of Solar Cycle 25 using Monthly Mean SSN					
Pred. 1	%13.13	0.9843	3.8	2030-05	10.5
Pred. 2	%15.11	0.9796	4.8	2030-07	10.67
Pred. 3	%12.01	0.9839	5.3	2030-07	10.67
Monthly Averaged Version of the Solar Cycle 25 Predictions					
Av. Pred. 1	-	-	11.4	2030-04	10.42
Av. Pred. 2	-	-	12.1	2030-04	10.42
Av. Pred. 3	-	-	9.9	2030-06	10.58

Table 2 Prediction results of the maximum and the minimum of Solar Cycle 26 and monthly averaged version of the prediction results.

Predictions	Amp. of Max. (SSN)	Time of Max. (yyyy-mm)	Amp. of Min. (SSN)	Time of Min. (yyyy-mm)	Cycle Length (years)
Predictions of Solar Cycle 26 using Monthly Mean SSN					
Pred. 1	181.5	2034-06	1.9	2040-10	10.5
Pred. 2	164.6	2034-09	7.8	2040-02	9.67
Pred. 3	150.6	2035-03	4.7	2040-10	10.5
Monthly Averaged Version of the Solar Cycle 26 Predictions					
Av. Pred. 1	146.2	2035-06	8.5	2040-11	10.67
Av. Pred. 2	142.1	2034-04	13.9	2040-07	10.33
Av. Pred. 3	137.4	2035-06	9.5	2040-08	10.25

the range 70 – 100 years. We, therefore, propose two likely versions for Solar Cycle 26: i) a cycle similar to Solar Cycle 20 with a clear double peak or ii) a cycle with slightly fluctuating flat peak with a peak value of 150.6 – 181.5 monthly mean sunspot numbers (or with 137.4 – 146.2 monthly averaged sunspot numbers), and overall structure similar to Solar Cycle 20.

Table 1 presents the results of the first part of our predictions (minimum of Solar Cycle 25). The table includes the successful prediction parameters, MAE and ρ values, cycle amplitudes in minimum, times of minimum and cycle lengths. For Solar Cycle 25, maximum amplitudes and times are not given, assuming that the maximum is already reached. In addition, a 13-month average is applied to the successful predictions in the top part of the table produced from the monthly mean SSN and the results are presented in the same format in the bottom part of the table. According to the 13-month average of the predictions, the minimum of Solar Cycle 25 is expected in April 2030 with monthly averaged sunspot numbers of 9.9 – 12.1 and a cycle length of around 10.42 years.

Table 2 presents the results of our predictions of Solar Cycle 26 similar to Table 1, which are a continuation of the same successful predictions. Since it is a continuation of the predictions in Table 1, the MAE and ρ values are the same and are not given here again. This table also includes the maximum amplitudes and times of Solar Cycle 26. According to the 13-month average of the predictions, maximum monthly averaged sunspot numbers of 137.4 – 146.2 is expected in June 2035 and minimum monthly averaged sunspot numbers of 8.5 – 13.9 is expected in November 2040, with a cycle length of around 10.67 years.

4. Dependence of Prediction Performance on the Library and Prediction Sets: Importance of the Split Point

Chaos theory studies dynamical systems that are sensitive to initial conditions. In chaotic systems, small changes in initial conditions may lead to significant variations of its future behavior. Thus, initial conditions are limiting to long-term predictability.

In the Simplex Projection method, precise determination of the initial conditions directly affects the performance of the prediction routine. The importance of the point where the prediction begins and the observed data set ends (starting point) was first demonstrated by Sarp et al. (2018).

The prediction performance does not only depend on the starting point of the prediction. In order to obtain the successful prediction, the point where the library set ends and the prediction set starts, i.e., where the sets are exactly split (which is called the split point) is extremely important for the prediction performance (see Section 2 for details).

Figure 6 shows an example of a successful (top left) and an unsuccessful (bottom left) prediction generated from monthly mean data, with the monthly averaged versions plotted in the right column. These predictions were produced using the same parameters under the same conditions (Pred. 1, Figure 5, $E = 2$, $\tau = 24$). The unsuccessful prediction notably underestimates the amplitude of the ascending branch of Solar Cycle 25, as well as all following phases of Solar Cycles 25 and 26 as compared to the observed data and the successful prediction shown in the top panels. The unsuccessful prediction was due to a change in the optimal split point.

The monthly mean data set used for the prediction consists of about 3250 data points covering the period from July 1749 to December 2019. For the successful prediction, the first 2197 points were taken as the library and the remaining 1053 points as the prediction set, while the split point for the unsuccessful prediction was taken at 2177/1073. In other words, the last 20 points of the library set were moved from the library to the prediction set. Successful Pred. 1 (top) matches well the beginning of the ascending branch of Solar Cycle 25 and consistently predicts the potential shape of Solar Cycle 26. Unsuccessful Pred. 1 (bottom), on the other hand, shows significant deviations in the ascending branch, and the continuation of the prediction and the potential shape of Solar Cycle 26 have major shifts. Furthermore, the MAE is 13.13% for the successful prediction and 23.49% for the unsuccessful prediction. Thus, deviations from the optimal split point resulted in nearly twofold increase of the MAE parameter. These two prediction results reveal the importance of the selection of the split point between library and prediction sets in nonlinear and chaotic systems.

Among the methods available for identifying optimal split points are grid search (Bergstra and Bengio 2012; Li et al. 2018), systematic trial-and-error procedures in time-series modeling (Box et al. 2015), and scanning or model selection strategies based on cross-validation (Stone 1974; Kohavi 1995). Furthermore, advanced approaches such as Bayesian optimization or Efficient Global Optimization (EGO) may also be employed for this purpose (Jones, Schonlau, and Welch 1998). In this study, predictions were produced by testing all candidate points across a wide range of the data set as split points (every point between 1000

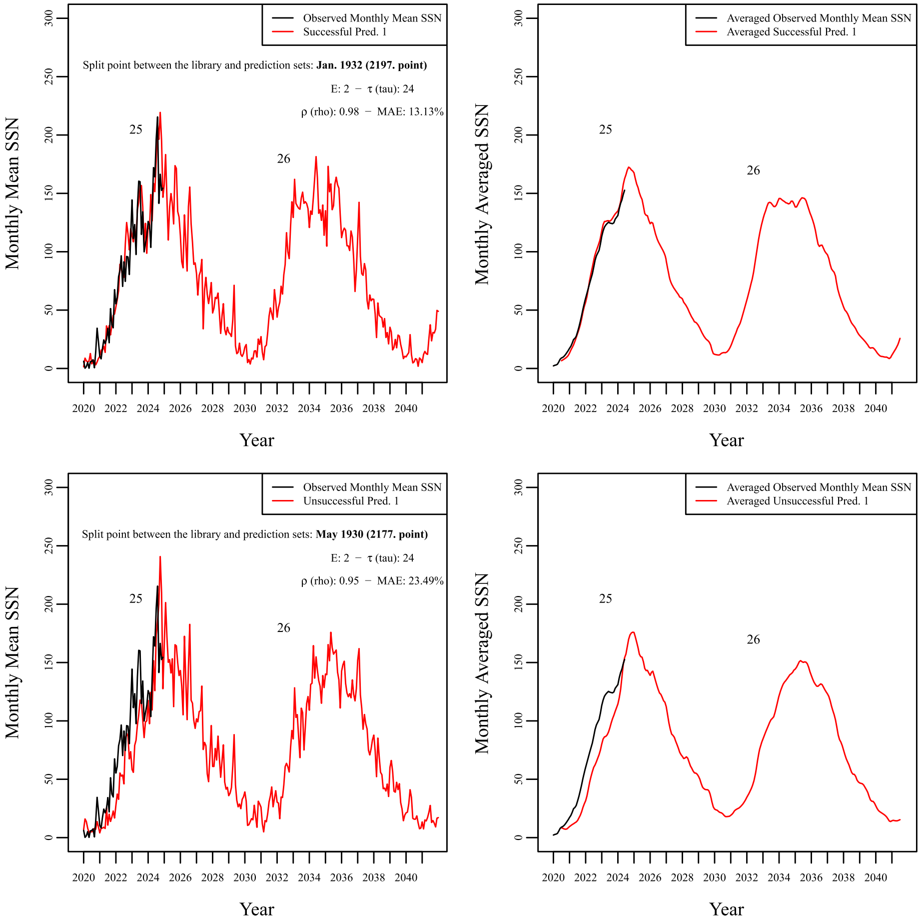

Figure 6 Predictions of Solar Cycles 25 and 26 generated using monthly mean sunspot numbers by the selection of optimal (top left) and non-optimal (bottom left) split point for the sets, and monthly averaged version of these predictions (top and bottom right).

and 2700) and using all combinations of E and τ for each candidate. Thus, the most successful predictions, those with the lowest MAE and highest ρ values, were used to determine the most appropriate split points.

5. Conclusions and Discussions

In this study, we applied the Simplex Projection method to the monthly mean SSN series to predict the remainder of Solar Cycle 25 and the next full Solar Cycle 26. Our findings are as follows:

1. Unsmoothed monthly mean data, which is not preferred for prediction studies in the literature, may better characterize the system dynamics.
2. The choice of the split point between the library and prediction sets is very important for the successful prediction of chaotic and nonlinear systems.

3. The minimum of Solar Cycle 25 is predicted to occur around May–July 2030 with monthly mean sunspot numbers reaching 3.8 – 5.3. A 13-month average was applied to these prediction results which changed the predictions for the cycle minimum to April–June 2030 with the minimal monthly averaged sunspot numbers of 9.9 – 12.1.
4. The maximum of Solar Cycle 26 is predicted to occur between June 2034–March 2035 with monthly mean sunspot numbers between 150.6 – 181.5, and the minimum will be reached in October 2040 with monthly mean sunspot numbers between 1.9 – 7.8. According to the 13-month average of the predictions, the maximum of Solar Cycle 26 expected around June 2035 with monthly averaged sunspot numbers of 137.4 – 146.2 and the minimum around July–November 2040 with monthly averaged sunspot numbers of 8.5 – 13.9.
5. Solar Cycle 26 is predicted to be weaker than Solar Cycle 25 but stronger than Solar Cycle 24. We suggest two possible shapes for Solar Cycle 26: a cycle similar to Solar Cycle 20 with a clear double peak or or a cycle with a flat peak that fluctuates slightly at its maximum.
6. The tendency of the Solar Cycle 26 prediction to resemble Solar Cycle 20 could be evidence of the Gleissberg Cycle, which has a period interval of about 70 – 100 years. Therefore, Solar Cycle 26 could be the end of a long-term period or the beginning of a new long-term period such as the Gleissberg Cycle.

Our study suggests that predictions based on unsmoothed monthly mean data may produce better results as compared to those based on smoothed data. Smoothed data is commonly used in contemporary studies utilizing various neural networks such as Neural Basis Expansion Analysis for interpretable Time Series forecasting (N-BEATS) and Long Short-Term Memory (LSTM). Moreover, precursor methods, AutoRegressive models (AR), parametric approaches such as ARIMA/SARIMA, Support Vector Regression (SVR), Random Forests (RF), and hybrid models also frequently utilize smoothed/filtered data to achieve better performance. However, our findings indicate that unsmoothed data can produce high-quality predictions, potentially capturing important short-term variations in solar activity that might be removed by filtering. The use of the Simplex Projection method contributes to the broader effort of refining solar cycle prediction models. While techniques like LSTM and hybrid models have shown promise, their reliance on smoothed data could reduce sensitivity to abrupt changes in solar activity. Our findings suggest reconsidering the over-reliance on data smoothing in predictions. We conclude that unsmoothed data can yield results comparable to or even better than smoothed data, potentially improving the performance of solar cycle prediction models.

Several studies have emphasized the importance of selecting appropriate initial points when predicting solar cycles, especially for the start point of prediction. Sarp et al. (2018), found that the prediction starting point is a crucial parameter in non-linear prediction schemes, identifying optimal points by analyzing previous cycles. Recently, Zhu et al. (2023) developed an optimized LSTM neural network model to predict Solar Cycle 25, emphasizing the significance of selecting optimal hyperparameters, including input length and prediction horizon (or time to prediction). These studies collectively underscore that the selection of initial and final points is critical for accurate predictions of chaotic and non-linear systems like solar cycles. In this study, we demonstrated that carefully selecting the split point between the library and prediction sets is crucial for successful performance of the Simplex Projection method. By analyzing starting and final points of previous cycles, we were able to select appropriate split points, leading to more accurate predictions/postcast compared to those presented in Sarp et al. (2018) study. Moreover, we further seriously im-

Table 3 Comparison of Solar Cycle 26 predictions by various methods for maximum amplitudes and times.

References	Methods	Amp. of Max. (Av./Sm. SSN)	Time of Max. (yyyy-mm)
This Study	*Simplex Projection*	*137.4 – 146.2*	*2035-06*
Tang (2015)	Grey Topological	130	2033-11
Singh and Bhargawa (2019)	Simplex Projection	78 ± 7	2036-06
Wu and Qin (2021)	TMLP & TMLP-E	107.3	2035-11
Kalkan, Fawzy, and Saygac (2023)	NARX	113.25	2036-10
Liu et al. (2023)	LSTM	135	2035-01
Cao et al. (2024)	Informer & Transformer	133.4	2036-06
Luo and Tan (2024)	Phase Similar	133 ± 3.2	2035 – 2036
Xu, Jain, and Xing (2024)	SARIMA + ML	141	2037-03
Rodríguez et al. (2024)	Multivariate ML	121.2 ± 10	2034-09

proved Sarp et al. (2018) prediction performance especially for the previous five cycles and we have almost perfectly postcasts (see Figure 2).

Sarp et al. (2018) study predicts the peak of the 13-month smoothed sunspot number for Solar Cycle 25 to reach 154 ± 12 and the cycle minimum to occur around mid-2030. Zhu et al. (2023) used an optimized LSTM model based on sunspot area data and predicted a monthly mean sunspot number to peak in January 2025 with a magnitude of about 213, indicative of a strong cycle, in line with our predictions of Solar Cycle 25 (see Figure 5, top panel). Penza et al. (2023) also presented predictions based on solar activity indicators (the core-to-wing ratio of Mg II at 280 nm, the solar radio flux at 10.7 cm (F10.7), and the total solar irradiance (TSI)), aligning their predictions with observations during the ascending phase of Solar Cycle 25, further supporting the reliability of predictions across different models. They highlighted consistency with the Gnevyshev–Ohl rule (Gnevyshev and Ohl 1948), which states that odd-numbered cycles are typically stronger than their preceding even cycles.

Wu and Qin (2021) used the Two-parameter Modified Logistic Prediction (TMLP) model and its Extension (TMLP-E) to predict Solar Cycles 25 and 26. They predicted the maximum of Solar Cycle 25 to occur in October 2024 with a low monthly smoothed SSN of 115.1, and the cycle minimum in January 2031. For Solar Cycle 26, they predicted a maximum with 107.3 monthly smoothed SSN in November 2035 and a minimum in August 2041. While the maximum amplitude results in the study are lower compared to our findings, the time of maximum/minimum are quite well matched. Kalkan, Fawzy, and Saygac (2023) used the Nonlinear AutoRegressive eXogenous (NARX) neural network approach to predict Solar Cycles 25 and 26. They expected a maximum with 116.6 monthly smoothed SSN in February 2025 and a minimum in 2030 for Solar Cycle 25, while for Solar Cycle 26, they found a maximum with 133.25 monthly smoothed SSN in October 2026 and a minimum in 2040. Cao et al. (2024) produced various predictions for unsmoothed and smoothed monthly SSN using informer and transformer models. They found that Solar Cycle 26 would be slightly weaker than Solar Cycle 25, which is stronger than Solar Cycle 24 with a double peak structure from the informer method. Based on the smoothed results, the first peak of Solar Cycle 26 is expected in February 2034 with 117.6 monthly smoothed SSN and the second peak is expected in May 2036 with 133.4 monthly smoothed SSN. We found the maximum amplitude of Solar Cycle 26 will be between 137.4 and 146.2 for monthly averaged/smoothed SSN, which is slightly different from the results of Kalkan, Fawzy, and

Saygac (2023) and Cao et al. (2024). But the maximum will occur in June 2035, which is different from the results of these studies. We found that Solar Cycle 26 will be weaker than Solar Cycle 25 and stronger than Solar Cycle 24 is in line with Cao et al. (2024). Table 3 provides a comparison of the results of Solar Cycle 26 predictions by various methods, including the results above and our study.

Based on analysis of the postcasts, predictions, amplitudes and maximum times presented in this study, we conclude that the double cycle prediction approach may generate successful predictions. The double cycle prediction allows us to examine the cyclic variations of solar activity beyond one cycle. The prediction of Solar Cycle 26 will be further improved once Solar Cycle 25 is completed and the library and the prediction sets can be extended farther in the future. These results are produced by a model-free empirical method and should be interpreted as statistical predictions; their skill is conditional on the degree of state dependence in the data and may decrease as stochastic variability increases.

Author Contributions All data preparation and analyses were performed by Kemalhan GERÇEKER under the supervision of Dr. Ali KILCIK. Dr. Vasyl YURCHYSHYN read the draft manuscript and made invaluable contributions to the current scientific version of the manuscript and to the correction of the English. Dr. Atila OZGUC read the draft manuscript and contributed to the final version of the manuscript.

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Data Availability The monthly mean total Sunspot number and the 13-month smoothed monthly total sunspot number data sets used in this manuscript are publicly available from the http://www.sidc.be/SILSO/home.

Declarations

Competing Interests The authors declare no competing interests.

References

Bergstra, J., Bengio, Y.: 2012, Random search for hyper-parameter optimization. *J. Mach. Learn. Res.* **13**, 281. https://jmlr.org/papers/v13/bergstra12a.html.

Bhowmik, P., Jiang, J., Upton, L., Lemerle, A., Nandy, D.: 2023, Physical models for solar cycle predictions. *Space Sci. Rev.* **219**, 40. DOI.

Box, G.E.P., Jenkins, G.M., Reinsel, G.C., Ljung, G.M.: 2015, *Time Series Analysis: Forecasting and Control*, 5th edn. Wiley, Hoboken. ISBN 9781118674925. DOI.

Bradley, E., Kantz, H.: 2015, Nonlinear time-series analysis revisited. *Chaos* **25**, 091102. DOI.

Cameron, R.H., Jiang, J., Schüssler, M., Gizon, L.: 2014, Physical causes of solar cycle amplitude variability. *J. Geophys. Res. Space Phys.* **119**, 680. DOI.

Cao, J., Xu, T., Deng, L., Zhou, X., Li, S., Liu, Y., Zhou, W.: 2024, An improved prediction of solar cycles 25 and 26 using the informer model: Gnevyshev peaks and North–South asymmetry. *Astrophys. J.* **969**, 120. DOI.

Charbonneau, P.: 2020, Dynamo models of the solar cycle. *Living Rev. Sol. Phys.* **17**, 4. DOI.

Charbonneau, P., Dikpati, M.: 2000, Stochastic fluctuations in a Babcock-Leighton model of the solar cycle. *Astrophys. J.* **543**, 1027. DOI.

Chowdhury, P., Sarp, V., Kilcik, A., Ray, P.C., Rozelot, J.P., Obridko, V.N.: 2022, A non-linear approach to predicting the amplitude and timing of the sunspot area in cycle 25. *Mon. Not. R. Astron. Soc.* **513**, 4152. DOI.

Crosson, I.J., Binder, P.M.: 2009, Chaos-based forecast of sunspot cycle 24. *J. Geophys. Res.* **114**, A1. DOI.

de Toma, G., White, O.R., Chapman, G.A., Walton, S.R., Preminger, D.G., Cookson, A.M.: 2004, Solar cycle 23: an anomalous cycle? *Astrophys. J.* **609**, 1140. DOI.

Deyle, E.R., Fogarty, M., Hsieh, C.H., Kaufman, L., MacCall, A.D., Munch, S.B., Sugihara, G.: 2013, Predicting climate effects on Pacific sardine. *Proc. Natl. Acad. Sci.* **110**, 6430. DOI.

Echer, E., Rigozo, N.R., Nordemann, D.J.R., Vieira, L.E.A.: 2004, Prediction of solar activity on the basis of spectral characteristics of sunspot number. *Ann. Geophys.* **22**, 2239. DOI.

Eddy, J.A.: 1976, The Maunder minimum: the reign of Louis XIV appears to have been a time of real anomaly in the behavior of the Sun. *Science* **192**, 1189. DOI.

Gleissberg, W.: 1939, A long-periodic fluctuation of the sun-spot numbers. *Observatory* **62**, 158.

Gnevyshev, M.N., Ohl, A.I.: 1948, On the 22-year cycle of solar activity. *Astron. Zh.* **25**, 18.

Gopalswamy, N., Michalek, G., Yashiro, S., Mäkelä, P., Akiyama, S., Xie, H.: 2023, What do halo CMEs tell us about solar cycle 25? *Astrophys. J. Lett.* **952**, L13. DOI.

Hathaway, D.H., Wilson, R.M., Reichmann, E.J.: 1994, The shape of the sunspot cycle. *Sol. Phys.* **151**, 177. DOI.

Jevtić, N., Schweitzer, J.S., Cellucci, C.J.: 2001, Research note: nonlinear time series analysis of northern and southern solar hemisphere daily sunspot numbers in search of short-term chaotic behavior. *Astron. Astrophys.* **379**, 611. DOI.

Jiang, J.: 2023, Nonlinear and stochastic mechanisms of the solar cycle and their implications for the cycle prediction. *Proc. Int. Astron. Union* **19**, 98. DOI.

Jones, D.R., Schonlau, M., Welch, W.J.: 1998, Efficient global optimization of expensive black-box functions. *J. Glob. Optim.* **13**, 455. DOI.

Kalkan, M.Y., Fawzy, D.E., Saygac, A.T.: 2023, Predictions of solar activity cycles 25 and 26 using non-linear autoregressive exogenous neural networks. *Mon. Not. R. Astron. Soc.* **523**, 1175. DOI.

Kantz, H., Schreiber, T.: 2003, *Nonlinear Time Series Analysis*, Cambridge University Press, Cambridge. DOI.

Kilcik, A., Anderson, C.N.K., Rozelot, J.P., Ye, H., Sugihara, G., Ozguc, A.: 2009, Nonlinear prediction of solar cycle 24. *Astrophys. J.* **693**, 1173. DOI.

Kilcik, A., Chowdhury, P., Sarp, V., Yurchyshyn, V., Donmez, B., Rozelot, J.P., Ozguc, A.: 2020, Temporal and periodic variation of the MCMESI for the last two solar cycles; comparison with the number of different class X-ray solar flares. *Sol. Phys.* **295**, 1. DOI.

Kohavi, R.: 1995, A study of cross-validation and bootstrap for accuracy estimation and model selection. In: *Proc. 14th Int. Joint Conf. Artif. Intell. (IJCAI'95)*, Morgan Kaufmann, Montreal, 1137. https://www.ijcai.org/Proceedings/95-2/Papers/016.pdf.

Lemerle, A., Charbonneau, P., Carignan-Dugas, A.: 2015, A coupled 2×2D Babcock-Leighton solar dynamo model. I. Surface magnetic flux evolution. *Astrophys. J.* **810**, 78. DOI.

Letellier, C., Aguirre, L.A., Maquet, J., Gilmore, R.: 2006, Evidence for low dimensional chaos in sunspot cycles. *Astron. Astrophys.* **449**, 379. DOI.

Li, L., Jamieson, K., DeSalvo, G., Rostamizadeh, A., Talwalkar, A.: 2018, Hyperband: a novel bandit-based approach to hyperparameter optimization. *J. Mach. Learn. Res.* **18**, 1. https://jmlr.org/papers/volume18/16-558/16-558.pdf.

Liu, X., Zeng, S., Deng, L., Zeng, X., Zheng, S.: 2023, Predicting the 25th and 26th solar cycles using the long short-term memory method. *Publ. Astron. Soc. Jpn.* **75**, 691. DOI.

Lorenz, E.N.: 2017, Deterministic nonperiodic flow 1. In: *Universality in Chaos*, 2nd edn. **367**, Routledge, London. DOI.

Luo, P.X., Tan, B.L.: 2024, Long-term evolution of solar activity and prediction of the following solar cycles. *Res. Astron. Astrophys.* **24**, 035016. DOI.

Maye, A., Hsieh, C.H., Sugihara, G., Brembs, B.: 2007, Order in spontaneous behavior. *PLoS ONE* **2**, e443. DOI.

McIntosh, S.W., Leamon, R.J.: 2017, Deciphering solar magnetic activity: spotting solar cycle 25. *Front. Astron. Space Sci.* **4**, 4. DOI.

McIntosh, S.W., Chapman, S., Leamon, R.J., Egeland, R., Watkins, N.W.: 2020, Overlapping magnetic activity cycles and the sunspot number: forecasting sunspot cycle 25 amplitude. *Sol. Phys.* **295**, 1. DOI.

Penza, V., Bertello, L., Cantoresi, M., Criscuoli, S., Berrilli, F.: 2023, Prediction of solar cycle 25: applications and comparison. *Rend. Lincei Sci. Fis. Nat.* **34**, 663. DOI.

Pesnell, W.D., Schatten, K.H.: 2018, An early prediction of the amplitude of solar cycle 25. *Sol. Phys.* **293**, 112. DOI.

Petrovay, K.: 2020, Solar cycle prediction. *Living Rev. Sol. Phys.* **17**, 2. DOI.

Rodríguez, J.V., Sánchez Carrasco, V.M., Rodríguez-Rodríguez, I., Pérez Aparicio, A.J., Vaquero, J.M.: 2024, Predicting solar cycle 26 using the polar flux as a precursor, spectral analysis, and machine learning: crossing a gleissberg minimum? *Sol. Phys.* **299**, 117. DOI.

Sarp, V., Kilcik, A., Yurchyshyn, V., Rozelot, J.P., Ozguc, A.: 2018, Prediction of solar cycle 25: a non-linear approach. *Mon. Not. R. Astron. Soc.* **481**, 2981. DOI.
Singh, A.K., Bhargawa, A.: 2019, Prediction of declining solar activity trends during solar cycles 25 and 26 and indication of other solar minimum. *Astrophys. Space Sci.* **364**, 12. DOI.
Singh, A.K., Siingh, D., Singh, R.P.: 2010, Space weather: physics, effects and predictability. *Surv. Geophys.* **31**, 581. DOI.
Spiegel, E.A.: 2009, Chaos and intermittency in the solar cycle. *Space Sci. Rev.* **144**, 25. DOI.
Stone, M.: 1974, Cross-validatory choice and assessment of statistical predictions. *J. R. Stat. Soc., Ser. B* **36**, 111. DOI.
Sugihara, G.: 1994, Nonlinear forecasting for the classification of natural time series. *Philos. Trans. R. Soc. Lond. A* **348**, 477. DOI.
Sugihara, G., May, R.M.: 1990, Nonlinear forecasting as a way of distinguishing chaos from measurement error in time series. *Nature* **344**, 734. DOI.
Sugihara, G., May, R., Ye, H., Hsieh, C.H., Deyle, E., Fogarty, M., Munch, S.: 2012, Detecting causality in complex ecosystems. *Science* **338**, 496. DOI.
Svalgaard, L.: 2013, Solar activity–past, present, future. *J. Space Weather Space Clim.* **3**, A24. DOI.
Takens, F.: 1981, Detecting strange attractors in turbulence. In: Rand, D.A., Young, L.-S. (eds.) *Dynamical Systems and Turbulence, Warwick 1980*, *Lecture Notes in Mathematics* **898**, Springer, Berlin, 366. DOI.
Tang, J.: 2015, Application of the Grey topological theory in the prediction of yearly mean sunspot numbers. *Chin. Astron. Astrophys.* **39**, 45. DOI.
Tsonis, A.A., Deyle, E.R., May, R.M., Sugihara, G., Swanson, K., Verbeten, J.D., Wang, G.: 2015, Dynamical evidence for causality between galactic cosmic rays and interannual variation in global temperature. *Proc. Natl. Acad. Sci.* **112**, 3253. DOI.
Wang, Z.F., Jiang, J., Wang, J.X.: 2025, Observation-based iterative map for solar cycles. I. Nature of solar cycle variability. *Astrophys. J.* **984**, 183. DOI.
Wang, X., Chen, Y., Toth, G., Manchester, W.B., Gombosi, T.I., Hero, A.O., Liu, Y.: 2020, Predicting solar flares with machine learning: investigating solar cycle dependence. *Astrophys. J.* **895**, 3. DOI.
Wu, S.S., Qin, G.: 2021, Predicting sunspot numbers for solar cycles 25 and 26. ArXiv preprint. arXiv. DOI.
Xu, Q., Jain, R., Xing, W.: 2024, Data-driven forecasting of sunspot cycles: pros and cons of a hybrid approach. *Sol. Phys.* **299**, 25. DOI.
Ye, H., Beamish, R.J., Glaser, S.M., Grant, S.C., Hsieh, C.H., Richards, L.J., Sugihara, G.: 2015, Equation-free mechanistic ecosystem forecasting using empirical dynamic modeling. *Proc. Natl. Acad. Sci.* **112**, E1569. DOI.
Zhu, H., Zhu, W., He, M.: 2022, Solar cycle 25 prediction using an optimized long short-term memory mode with F10.7. *Sol. Phys.* **297**, 157. DOI.
Zhu, H., Chen, H., Zhu, W., He, M.: 2023, Predicting solar cycle 25 using an optimized long short-term memory model based on sunspot area data. *Adv. Space Res.* **71**, 3521. DOI.